

Himanshu Pandey

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PROFESSIONAL SUMMARY

Data Science and ML Engineer with strong foundation in Python, Machine Learning, and MLOps. Experienced in building end-to-end ML pipelines, time-series forecasting, and algorithmic trading systems. Passionate about designing scalable data-driven solutions with solid understanding of statistical modeling and production deployment.

EDUCATION

- Indian Institute of Technology- Madras** Chennai, India
 - Bachelor of Science - Data Science and Applications* Sept 2023 - Aug 2026
 - Relevant Coursework:* Data Structures, Machine Learning, Database Management Systems, Statistics

SKILLS SUMMARY

- **Languages:** Python, SQL, JavaScript
- **Machine Learning:** Scikit-learn, Feature Engineering, Model Evaluation, Supervised & Unsupervised Learning
- **Data:** Pandas, NumPy, EDA, Data Visualization, ETL
- **MLOps:** FastAPI, Docker, Kubernetes, CI/CD, Model Deployment
- **Web & Tools:** Flask, REST APIs, React, MongoDB, Git, Linux, Jupyter, VS Code

EXPERIENCE

- Shodhshala** Remote
 - Data Science Intern – ML* Jan 2026 – Present
 - **Feature Engineering & Model Performance:** Engineered 30+ features from raw market data, improving model signal quality by ~15% and overall strategy performance.
 - **Algorithmic Strategy Development:** Developed and backtested algorithmic trading strategies using statistical methods, improving average win rate by ~12%.
 - **Database Management:** Managed PostgreSQL databases handling 1M+ rows of high-frequency financial data with optimized query performance.
 - **ETL Pipeline Automation:** Built automated ETL pipelines reducing data preprocessing time by ~40% across large-scale trading datasets.
 - **Production Monitoring:** Monitored deployed model performance and delivered actionable insights to refine live trading strategies.

PROJECTS

- Movie Audience Prediction — ML & Time Series Forecasting**
 - Tech: Python, XGBoost, LightGBM, Scikit-learn, Pandas*
 - **Model Development:** Built an end-to-end audience forecasting system using lag features (1, 7, 14, 30-day) and rolling means with Random Forest, XGBoost, and LightGBM; selected best model via RMSE comparison.
 - **Ensemble Engineering:** Engineered a weighted ensemble blending day-of-week patterns, 7 & 30-day trends, and theater-level behavior with multi-level fallback for improved prediction stability over single-model baseline.
- E-commerce RTO Prediction — Classification & Feature Engineering**
 - Tech: Python, Gradient Boosting, Scikit-learn, Pandas*
 - **Classifier Performance:** Built a Gradient Boosting classifier achieving 68.75% F1-Score and 73.33% ROC-AUC to predict Return-to-Origin orders; benchmarked 5 models using cross-validation (CV: 70.66% $\pm$ 1.42%).
 - **Feature Engineering:** Engineered 4 domain-specific features (Cost Weight Ratio, Weight Bin, etc.) contributing 54%+ of total feature importance, with Weight and Discount identified as top RTO drivers.
- Customer Purchase Value Prediction — Regression Pipeline**
 - Tech: Python, XGBoost, Scikit-learn, Pandas, NumPy*
 - **Pipeline & Validation:** Built an end-to-end regression pipeline engineering 20+ behavioral, temporal, and user-level features; used GroupKFold validation grouped by userId to prevent data leakage.
 - **Model Performance:** Trained XGBoost ($R^2 = 0.422$) outperforming HistGradientBoosting baseline ($R^2 = 0.374$) by ~13% on a Kaggle competition dataset.
- Vehicle Parking System — Full-Stack Distributed Web Application**
 - Tech: Python, Flask, Vue.js, SQLite, SQLAlchemy, Celery, Redis, REST APIs*
 - **Full-Stack Development:** Designed a role-based parking management platform (Admin & User) with real-time slot booking, reservation history, and dedicated dashboards via REST API integration.
 - **Async Job Processing:** Implemented Celery and Redis for automated daily reminders and monthly CSV report generation delivered via email using scheduled background tasks.
 - **Data Integrity:** Built soft-deletion logic preserving historical reservation records when parking lots or spots are removed from the system.

CERTIFICATIONS

- Diploma In Programming
- Diploma In Data Science
- Advanced Certificate in Programming Application Development